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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

LAST REVISION ON June 22, 2015





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 1

Introduction

Learning outcomes

Upon completing this lesson, you should be able to:

- *Define GIS*
- *Describe components of a GIS*
- *Distinguish different data types*

1.1. Introduction

GIS is an information system which is geographic in nature. It is a computer-based tool for mapping and analyzing geographic phenomenon that exist, and events that occur on Earth. It is



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a means of storing, retrieving, sorting, and comparing spatial data to support some analytic process. GIS technology integrates common database operations such as query and statistical analysis with the unique visualization and geographic analysis benefits offered by maps. It links graphical features (entities) to tabular data (attributes). These abilities distinguish GIS from other information systems and make it valuable to a wide range of public and private enterprises for explaining events, predicting outcomes, and planning strategies. GIS performs tasks faster and with more sophistication than traditional methods. GIS is a collection of computer hardware and software.

A GIS can be made up of a variety of software and hardware tools and geographic data for capturing, managing, analyzing, and displaying forms of geographically referenced information.



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The important factor is the level of integration of these tools to provide a smooth operating and fully functional geographic data processing environment by manipulating and managing digital spatial (geographic) and related attribute data.

In general, a GIS provides facilities for data capture, data management, data manipulation and data analysis, and the presentation of results in both graphic and report form, with a particular emphasis upon preserving and utilizing inherent characteristics of spatial data. The ability to incorporate spatial data, manage it, analyze it, and answer spatial questions is the distinctive characteristic of geographic information systems. Geographic location is an important attribute of activities, policies, strategies, and plans. Geographic problems involve an aspect of location, either in the information used to solve them, or in the



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solutions themselves.

As hardware, a GIS consists of a computer, memory, storage devices, scanners, printers, global positioning system units, and other physical components. If the computer is situated on a network, the network can also be considered an integral component of the GIS because it enables sharing of data and information that the GIS uses as input and creates as outputs.

As a tool, a GIS permits maintaining, analyzing, and sharing wealth of data and information. From the relatively simple task of mapping the path of a hurricane to the more complex task of determining the most efficient garbage collection routes in a city, a GIS is used across the public and private sectors. Online and mobile mapping, navigation, and location-based services are also personalizing and democratizing GISs by bringing maps and



mapping to the masses.

1.1.1. Advantages of GIS

- GIS explores both geographical and thematic components of data in a holistic way
- It stresses geographical aspects of a research question
- It allows handling and exploration of large volumes of data
- It allows integration of data from widely disparate sources
- It allows analysis of data to explicitly incorporate location
- It allows a wide variety of forms of visualization.

1.1.2. Disadvantages of GIS

- Geographical data is expensive



- The learning curve on GIS software can be long
- GIS shows spatial relationships but does not provide absolute solutions
- GIS originated from the earth sciences and computer science. Solutions may not be appropriate for humanities research

1.2. Approaches to GIS

In addition to recognizing the many definitions of a GIS, there are three general and overlapping approaches to understanding GISs

1. the application approach,
2. the developer approach,



3. the science approach.

Though most GIS users would probably identify with one approach more than another, they are not mutually exclusive. Moreover, as GISs and, more generally, information technology advance, the above approaches will be transformed and reshaped accordingly.

The application approach to GISs considers a GIS primarily to be a tool. From this perspective, a GIS is used to answer questions, support decision making, maintain an inventory of geographic data and information, and make maps. As a tool, there are certain skills that should be acquired and required in order to use and apply a GIS properly. The application approach to a GIS is more concerned with using and applying GISs to solve problems than the GIS itself.



In light of the rapid rate of technological and GIS innovation, in conjunction with the widespread application of GISs, new questions about GIS technology and its use are continually emerging. One of the most discussed topics concerns privacy, and in particular, what is referred to as locational privacy. Questions concerning locational privacy are of relatively little concern. However, with the advent of GPS and its integration into cars and other mobile devices, questions, debates, and even lawsuits concerning locational privacy and who has the right to such information are rapidly emerging.

1.2.1. GIS Examples

- ArcView
- ArcGIS



- MapInfo

1.3. Characteristics of GIS

- GIS integrates spatial data and other kinds of information within one system: it offers a consistent framework for analyzing space
- GIS makes connections between activities based on spatial proximity
- GIS provides the mechanisms for undertaking the manipulation and display of geographic knowledge

1.4. Functions/Subsystems of GIS

A GIS has four main functional subsystems. These are:



- a data input subsystem;
- a data storage and retrieval subsystem;
- a data manipulation and analysis subsystem;
- a data output and display subsystem.

1. **Data Input Assembly:** Before geographic data can be used in a GIS, the data must be converted into a suitable digital format. The process of converting data from paper maps into computer files is called digitizing. Modern GIS technology can automate this process fully for large projects using scanning technology; smaller jobs may require some manual digitizing. Today many types of geographic data already exist in GIS-compatible formats. Standard digital data exists in useable formats and can

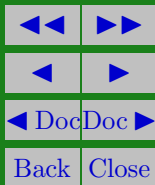


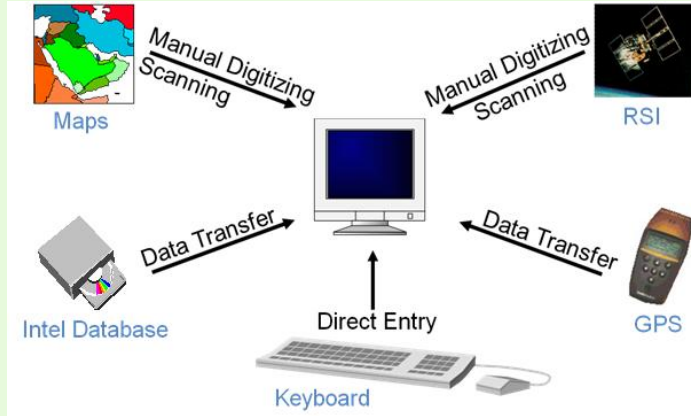
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usually be loaded directly into a GIS.

A data input subsystem allows the user to capture, collect, and transform spatial and thematic data into digital form.

The data inputs are usually derived from a combination of hard copy maps, aerial photographs, remotely sensed images, reports, survey documents, etc.





2. **Data Storage:** The data storage and retrieval subsystem organizes the data, spatial and attributes, in a form which permits it to be quickly retrieved by the user for analysis, and permits rapid and accurate updates to be made to the database. This component usually involves use of



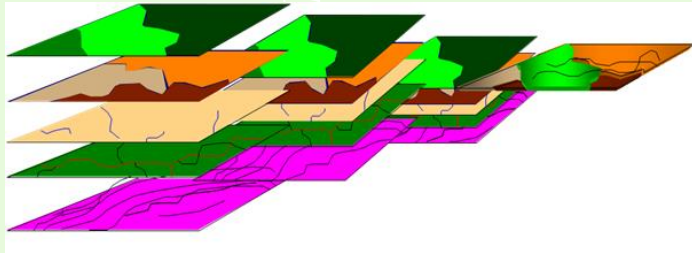
a database management system (DBMS) for maintaining attribute data. Spatial data is usually encoded and maintained in a proprietary file format. For small GIS projects it may be sufficient to store geographic information as simple files. There comes a point, however, when data volumes become large and the number of data users becomes more than a few, that it is best to use a database.

3. **Data Analysis and Manipulation:** This is one the most powerful functions in a GIS. It gives the user the ability to place the cartographic representation of thematic data over another. The data manipulation and analysis subsystem allows the user to define and execute spatial and attribute procedures to generate derived information. This subsystem is commonly thought of as the heart of



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a GIS, and usually distinguishes it from other database information systems and computer-aided drafting (CAD) systems. It is likely that data types required for a particular GIS project will need to be transformed or manipulated in some way to make them compatible with your system. GIS technology offers many tools for manipulating spatial data and for weeding out unnecessary data.



4. **Data Output:** There are many forms of output for a GIS.



The first is usually a soft copy. A full functioning GIS can do this. It can plot/print given the equipment, the 2D and 3D perspective views on a 2D map. It entails Tables, Maps, Interactive Displays and 3-D Perspective View. The data output subsystem allows the user to generate graphic displays, normally maps, and tabular reports representing derived information products.

1.5. GIS Process

The GIS process involves six steps that are common to what is referred to as the end to end map-making process.

1. Define the spatial problem/question.
2. Define the GIS criteria.
3. Import or create the data sets



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4. Perform the GIS analysis
5. Create the output
6. Decide whether or not the output solves or answers the spatial problem/question. If not, then refine the problem and start the process again.





1.6. Components of a GIS

An operational GIS has a series of components that combine to make the system work. These components are critical to a successful GIS. A working GIS integrates five key components:

1. **Hardware:** Hardware is the computer system on which a GIS operates. Today, GIS software runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.
2. **Software:** GIS software provides the functions and tools needed to store, analyze, and display geographic information.
3. **Data:** The most important component of a GIS is the



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data. Geographic data and related tabular data can be collected in-house, compiled to custom specifications and requirements, or occasionally purchased from a commercial data provider. A GIS can integrate spatial data with other existing data resources, often stored in a corporate DBMS. The integration of spatial data (often proprietary to the GIS software), and tabular data stored in a DBMS is a key functionality afforded by GIS.

4. **People:** GIS technology is of limited value without the people who manage the system and develop plans for applying it to real world problems. GIS users range from technical specialists who design and maintain the system to those who use it to help them perform their everyday work. The identification of GIS specialists versus end users



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is often critical to the proper implementation of GIS technology.

5. **Methods:** A successful GIS operates according to a well-designed implementation plan and business rules, which are the models and operating practices unique to each organization.

As in all organizations dealing with sophisticated technology, new tools can only be used effectively if they are properly integrated into the entire business strategy and operation. To do this properly, it requires not only the necessary investments in hardware and software, but also in the retraining and/or hiring of personnel to utilize the new technology in the proper organizational context. Failure to implement your GIS without regard for a proper organizational commitment will result in an unsuc-



cessful system.

1.7. GIS Data Types

The basic data type in a GIS reflects traditional data found on a map. Accordingly, GIS technology utilizes two basic types of data which are stored for each item in the database. These are:

1. **Spatial data:** This data describes the absolute and relative location of geographic features. It says where the feature is co-ordinate based; Vector data and discrete features which are :
 - Points
 - Lines
 - Polygons (zones or areas): – Raster data:
 - A continuous surface



2. **Attribute data:** This data describes characteristics of the spatial features. These characteristics can be quantitative and/or qualitative in nature. Attribute data is often referred to as tabular data. Eg. statistics, text, images, sound, etc.

The coordinate location of a forestry stand would be spatial data, while the characteristics of that forestry stand, e.g. cover group, dominant species, crown closure, height, etc., would be attribute data. Other data types, in particular image and multimedia data, are becoming more prevalent with changing technology. Depending on the specific content of the data, image data may be considered either spatial, e.g. photographs, animation, movies, etc., or attribute, e.g. sound, descriptions, narration's, etc.



1.8. Classification of GIS

Different classification schemes of commercial GIS are proposed in literature. It is convenient to classify the main GIS software packages into six groups, based on their functionality and type:

1. **Professional:** Refers to fully featured software that has the capabilities of collection and editing GIS data database administration advanced and editing GIS data , database administration , advanced geoprocessing and analysis, and other specialist tools. Examples of professional GIS include ESRI ArcInfo and Smallworld GIS.
2. **Desktop:** This category is based on the desktop computer, with the focus on data use rather than data cre-



ation. It has excellent tools for making maps, reports, and charts. Most of the desktop GIS software have all adopted the Microsoft standards for interoperability and user interface. Examples include Autodesk World, ESRI ArcView, Intergraph GeoMedia, and MapInfo Professional, Quantum GIS

3. **Hand-held:** Hand-held GIS are lightweight systems designed for mobile and field use Have capabilities similar to hand held computers, supporting many display, query, and simple analytical applications, even on displays of 320 X 240 pixels displays of 320 X 240 pixels
4. **Component:** These are basically tool kits of GIS functions (components) that a reasonably knowledgeable programmer can use to build a full GIS software system.



These type of components can be used to create highly customized and optimized applications that can either be stand- alone or can be embedded within other software systems alone or can be embedded within other software systems. Examples include ESRI MapObjects, MapInfo MapX, and Blue Mable Geographics GeoObjects.

5. **Viewer:** It is a free GIS software that is able to display and query popular file formats. The GIS viewers have limited functional capabilities, restricted to display query and simple mapping display query and simple mapping. They do not support editing, sophisticated analysis, modelling or customization. Examples include ESRI's ArcReader, ArcExplorer now ArcGIS Explorer, Intergraph's GeoMedia Viewer and Mapinfo's ProViewer. The ratio-



nale of the viewers is to establish market share, and specific vendor terminology and data formats as de facto standards

6. **Internet:** These products harness the unique characteristics of the world wide by integrating GIS technology with web browsers and servers, and uses the hypertext transmission protocol (http) for communication. communication. Examples of Internet GIS products include Autodesk MapGuide, ESRI ArcIMS, Intergraph GeoMedia Web Map and MapInfo MapXtreme. ArcIMS, Intergraph GeoMedia Web Map and MapInfo MapXtreme.

Three other types of GIS software systems are: CAD-based GIS, Raster-based GIS, and GIS application servers.



1.9. Spatial Data

Spatial data refers to the real-world geographic objects of interest, such as streets, buildings, lakes, and countries, and their respective locations. In addition to location, each of these objects also possesses certain traits of interest, or attributes, such as a name, number of stories, depth, or population. GIS software keeps track of both the spatial and attribute data and permits us to link the two types of data together to create information and facilitate analysis.

Spatial data is essentially data with a location. It contains information about the location and shape of, and relationship among geographic features usually stored as coordinates. Spatial data comes in two types, VECTOR and RASTER. Geographic information systems work with two fundamentally dif-



ferent types of geographic models—the “vector model” and the “raster model.”

1.9.1. What makes data spatial?

Spatial data has particular characteristics. These can be described in terms of: shape, place and relationship to other spatial data (or geometry, location, and topology. It is also necessary to model real world data (such as a road or building) in terms of a geographical representation. For example, a road could be represented as a line and the building perhaps as a small box on a map. These features (line, box) are in fact models of the actual real world features. Sometimes these models are described as objects or entities too.

Another important aspect of spatial data is that it often



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contains attribute information. That implies that a description of the feature (the road) is held in some form. The description might be the name or the type of road (A, B,). This information might be held in a database record or simply written or depicted on a map.

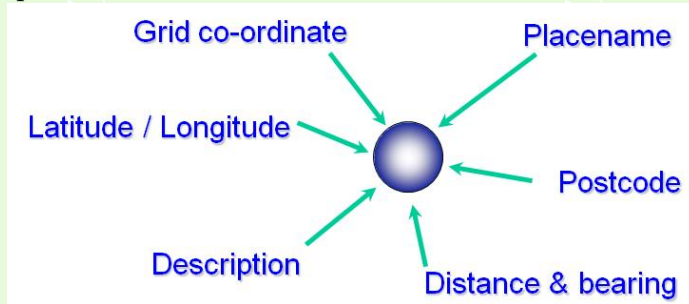
Finally spatial data by its very nature implies that relationships are also recorded. When we look at map data, we automatically interpret the relative locations of the spatial data. Computers require more explicit descriptions.

Spatial data thus refers to information that is associated with a location or place. It may be recorded on a map, held as records in a database or even be represented as a photograph. Remember that Geography is, in fact, the study of spatial information and that we are surrounded by geography. You will also dis-



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cover that most information is either spatial or has a spatial component.



1.10. Characteristics of spatial data

1. Location: This can be expressed in many ways. It may, however, differ in terms of:
 - Method e.g. co-ordinate, place name, Post Code



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- Precision e.g. $<1.0\text{m}$, City name, Post Code sector
 - Accuracy e.g. centre of building, front door position
2. Geometry: The shape of a building or county, the course of a river, the route of a road and the shape of the landscape / relief
 3. Topology: This term expresses how geographical data about one location relates to data about another. These concepts often include descriptions such as:
 - Connected to
 - Within
 - Adjacent to
 - North of . . .



1.10.1. Examples of Spatial Data

- Socio-economic data: It is often from national and local government, and is usually the product of population surveys and censuses. This data is also used by a number of commercial vendors who combine census information with other datasets to produce neighbourhood profiles classifying particular areas for marketing purposes. This ability to recognize particular markets based on geographical datasets is known as Geodemographics and is one of the fastest growth areas within GIS.
- Environmental Data: The collection and analysis of information about the environment was one of the driving forces behind the development of GIS and continues to be



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an important application area. Environmental data sets often tend to be large and require considerable management. Data often includes boundaries between vegetation types, for example, which are fuzzy i.e. they are not defined by a simple line. Conversely, socio-economic data is usually related to administrative boundaries, which are sharp if artificial.



Revision Questions

EXERCISE 1.  Define GIS.

Example . Describe the characteristics of a GIS that distinguish it from other kinds of computerized systems.

Solution: Revise. ☐

EXERCISE 2.  Describe your current location in terms of name of building, Postcode, Grid reference & Latitude/Longitude.

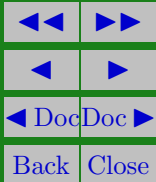


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Solutions to Exercises

Exercise 1. Revise.

Exercise 1





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Exercise 2. *Revise.*

Exercise 2

